The Theory of Closed Native Verification Systems

Version 3.0

This document constitutes a more rigorous formalization attempt of the CNVS Theory presented in the previous documentation.
Additional documents are expected to be released that will replace or supplement the present work.

Author: Massimo Comitato

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1. Introduction

This document constitutes the formalization of a theoretical system proposed as an Ideal Model for the verification attestation of physical evidence of the existence of an object external to the system itself, starting from declarations made within the system.

In this document, the term *Object* is intended in its broadest sense as any reality that can be supported by a specific, unique, and objectively identifiable physical observation.

In summary, this theoretical model attempts to reconcile a specific truth claim with the objective verification of its consistency within physical reality.

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2. Abstract

This document defines a new class of systems called **Closed Native Verification Systems (CNVS)**.

In such systems, the existence, validity, and evolution of state are determined exclusively by internal verification processes supported by externally generated verification evidence, randomly distributed and ultimately governed by deterministic invariants.

Although objective verification processes remain external, their distribution—namely, the specific structure imposed by the system—reduces trust to the execution of a strict procedural framework that limits executor discretion through the progressive reduction of local and global operational knowledge.

The theory assumes three fundamental axioms that are imposed as necessary conditions for classifying a system as a Native Verification System.

The axioms define:

- the structural properties of the entire system;
- the emergent behavior related to security and verification dynamics;
- **the principles governing both external and internal processes.**

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3. Problem Statement

In traditional systems addressing the same problem:

- the existence of a state is a unilateral decision made by a governing authority (total trust);
- verification is entrusted to external auditors, generally constrained only by global rules with limited local references;
- the existence of state is admitted natively through issuer authority;
- state verification follows the declaration of existence;
- state is typically unitary;
- internal security decreases as adoption and the number of states increase, because the attack surface expands.

In a CNVS, the system reverses this logic:

- the existence of a state is not imposed by a governing authority, but is the result of an internal mathematical process;

- verification is entrusted to external verifiers, rigidly guided by a distribution model that uses restricted access to information as a method of collusion control;
- externally issued evidence must first pass strict internal consistency checks;
- state is not unitary;
- security emerges as a system property and increases as adoption grows.

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4. General Definitions

In this document, a System is defined as a structured set of objects and elements, regardless of type, possessing specific correlations and governed by preliminary assumptions that define its functions.

A System is considered C**losed** when no external intervention can alter its structure or produce outcomes that contradict its governing assumptions.

A System is defined as **Native Verification-Based** because the existence of all constituent elements and objects is admitted only after objective verification followed by a consistency check of the resulting attestations.

In other words:

> No element may come into existence unless it has first been verified.

A Native Verification System is defined as closed because no unauthorized external intervention can alter its structure or produce results that contradict its imposed assumptions.

For the purposes of the CNVS model, information (I_x) is defined as any finite ordered binary representation admitted by the System. Physical objects, measurements, attestations, certificates, signatures, and attributes are processed only through their Encoded Representation as elements of {0,1}*.

The System does not directly process physical Objects, but operates exclusively on their verifiable digital representation.

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5. Special Definitions

5.1 State and Structure

Let $E(t)$,

be the complete Set of all Structured Elements that compose the System at time t,
with $s_i \in E(t)$

Let $R(t)$,
be the Relations among the Structured Elements at time t,
with

$R(t) \subseteq E(t) \times E(t)$

Let C,
be the Deterministic Invariants imposed by the System, namely

$C = \{c_1, c_2, \ldots, c_m\}$

Let $V_{\text{(local or global)}}$,

be the Internal Verification Function (local or global) for the Determination of Logical consistency before the declaration of the Existence of State.

A State will therefore be defined as:

$$S(t) = [E(t), R(t), C, V_{Local}] \quad \textbf{(1)}$$

Since Structured Elements are not atomic, but may be fragmented randomly without losing their Characteristics and Properties,
thus rendering the System Recursive, we will therefore have:

$E(t) = \{s_1, s_2, s_3, \ldots, s_i\}$

with

$s_i = (id_i, D_i, At_i, Val_i, P_i)$ **(2)**

where,

$\forall\, id_i, \forall\, D_i \in (\{0,1\}^*)^z$

$\forall\, At_i \in (\{0,1\}^*)^x$

$\forall\, Val_i \in (\{0,1\}^*)^y$

$\forall\, P_i \subseteq \mathbb{P}$

$\mathbb{P}$ is the finite or implementation-defined set of logical properties admitted by the system,

with $Z, X, Y \in \mathbb{N}$.

Therefore,

id_i = unique identifier,

D_i = data (e.g., 2 grams, 2 parts, or 2 mL),

At_i = Attributes (purity, internal state, timestamp, turbidity, etc.),

Val_i = Validations (audits, certificates, signatures, etc.) expressing Validated,

$\mathbb{P}$ = Properties,
{Valid, Transferable, Locked, Verified, Suspended, Redeemable,…}

with

$P_i = f(D_i, At_i, Val_i)$, or more precisely:

$P_i = V_{Local}(D_i, At_i, Val_i)$. **(3)**

Fundamental Property:

<u>decomposition does not change type</u>.

$s_{ij} \sim s_i$

Explicitly, we will have:

$s_{ij} = (id_{ij}, D_{ij}, At_{ij,} Val_i, P_{ij})$

with $\forall$ n, $\forall$ i, $\forall$ j $\in \mathbb{N}$

that is, the same and identical Structural Attributes,
Same nature,
Same schema.

Therefore:

$\exists \{s_{i1},...,s_{ik}\} \subseteq E(t)$

i.e.:

$s_{i1}, s_{i2}, s_{i3}, ..., s_{ij} \subseteq E(t)$.

Hence

$$\forall s_i \in E(t), \exists \{s_{ij}\} \subseteq E(t) \qquad \textbf{(4)}$$

such that:

$$\text{structure}(s_{ij}) = \text{structure}(s_i) \qquad \textbf{(5)}$$

Decomposition is <u>Finite</u> at every level.

Furthermore, during the Decomposition process, uniformity is not necessarily maintained, such that the obtained sub-elements preserve equal portions of the original Structured Element, but the Distribution of the Decomposition may be non-uniform.

Non-uniform Decomposition among Fragments of the same Level does not affect the Conservation of Information, but certainly affects the technical operation of its Reconstruction.

Formally:

$\forall s_i \in E(t), \forall n \in \mathbb{N}$, $\exists$ a decomposition of depth n **(6)**

or more precisely

$\forall s_i \in E(t), \forall\, n \in \mathbb{N}, \exists \{s_i^{(1)}, s_i^{(2)}, \ldots, s_i^{(n)}\}$ **(7)**

such that

level 0 of Origin

$s_i^{(0)} = \{s_1^{(0)}, s_2^{(0)}, \ldots, s_i^{(0)}\}$

level 1

$Sub(s_i^{(0)}) = \{s_{i1}^{(1)}, s_{i2}^{(1)}, \ldots, s_{ij}^{(1)}\}$

level 2

$Sub(s_{ij}^{(1)}) = \{s_{ij1}^{(2)}, s_{ij2}^{(2)}, \ldots, s_{ijl}^{(2)}\}$

with $\forall\, n, \forall\, i, \forall\, j, \forall\, l \in \mathbb{N}$

Therefore, for every n there exists a subsequent Decomposition level such that:

$s_i^{(n+1)} \in Sub(s_i^{(n)})$ **(8)**

and each $s_i^{(n)}$ preserves the defined Structure of the original Elements.

Explicitly:

$s_{ij} = (id_{ij}^{(1)}, D_{ij}^{(1)}, At_{ij}^{(1)}, Val_i^{(1)}, P_{ij}^{(1)})$.

Schematically, the sequence of Decomposition levels will be:

$s_i \rightarrow s_i^{(1)} \rightarrow s_i^{(2)} \rightarrow \ldots \rightarrow s_i^{(n)}$

where:

$s_1^{(0)} = s_1$ (initial fragment of level zero)

$s_1^{(1)} =$ fragments of $s_1^{(0)}$

$s_1^{(2)} =$ fragments of $s_1^{(1)}$.

Therefore:

$\forall\, n \in \mathbb{N},\ \exists\, s_i^{(n)} \in E(t)$ **(9)**

Within each sub-level, the portion of the Element from the preceding Level is not distributed uniformly.

In $s_{ij}^{(1)}$, the notation j as a subscript referring to subsequent Decomposition levels for the same fragment of Order i is replaced by the superscript (1), which assumes the same meaning as j.

In the formulae described from point (5) to point (6), both notations are reported.

N.B. In this Document the First Level of Structural Decomposition (Level 0) is called Nodal and the Structured Fragments si(0) of the Nodal Level are called Nodes.

5.2 Relations

Let

$R(t) \subseteq E(t) \times E(t)$ **(10)**

where,

$E(t) \times E(t)$

represents the Cartesian product of all possible pairs of the Structured Elements of Set E at the same time t, where R(t) is given by the

Directional Relations recognized by the System. If $E(t) = \{s_1, s_2\}$,

therefore

$E(t) \text{ x } E(t) = \{(s_1, s_1), (s_1, s_2), (s_2, s_1), (s_2, s_2)\}$.

The System does not necessarily provide for all possible Relations; only some may be permitted.
For example

let

$R(t) = \{(s_1, s_2), (s_2, s_2)\}$

The Directional Relation could only exist between
the pair s_1 and s_2
and between
the pair s_2 and s_2,
which would therefore only be in relation to itself.
The System therefore does not fully recognize all possible pairs of the Cartesian product E x E, but only some.
It is the Specific Implementation of the System, as well as the Structural Definition of its Constituent Elements, that decides on the Relations.
There are no Constraints Imposed by the System on the Relations, other than those of Logical Consistency to the Invariants and to the Global Structure of the System itself.

In the System

The Relation is always <u>Directional</u> (non-symmetric),
being the fundamental Order for Verification,
such that

$(s_j, s_i) \neq (s_i, s_j)$.

Put another way,,

let

$\forall R(S_{i,} S_j) \in R(t)$,

does not imply that

$R(S_{j,} S_i) \in R(t)$,

therefore

$(s_i, s_j) \in R(t) \nRightarrow (s_j, s_i) \in R(t)$ **(11)**

All Relations allowed in the System must be
Logical,
Structural,
necessarily Directional (non-symmetric),

and non-Quantitative.
Relations are therefore Directional, and Connections are not weighted between Structured Elements.
The non-uniform Decomposition of Fragments does not change the Determination of the possible Relations between those Fragments at the same level of Decomposition.

The pair $(E(t),R(t))$ thus defines a directed graph over the structured elements of the system.

In this representation:

- each structured element $s_i \in E(t)$ corresponds to a vertex (node);
- each admissible relation $(s_i, s_j) \in R(t)$ corresponds to a directed edge;
- decomposition levels define a hierarchical structure in which nodes may generate descendant nodes through recursive fragmentation;
- nodal elements represent first-level vertices;
- fragmental elements represent vertices at deeper levels of decomposition.

The resulting Graph therefore captures both the logical relations among elements and the recursive structure of state decomposition.

5.3 Attributes (At $_i$)

Let

$$At_i = \{a_{i1}, a_{i2}, a_{i3}, \ldots, a_{ik}\} \qquad \textbf{(12)}$$

Attributes are observable data, not logical.

The Attribute is therefore an Objective Description of the Element.
Attributes may exhibit Internal Dependencies, but these cannot be modeled as Relationships within the System and are defined by the specific Implementation.

5.4 Property (P_i)

Properties are logical conditions that act on the single Structured Element, such that

$$P_i = f(D_i, At_i, Val_i) = V_{Local}(D_i, At_i, Val_i) \quad \textbf{(13)}$$

For example, a Property can be
valid, transferable, locked, verified, etc.
Properties are not data, but rather Evaluations that depend exclusively on the System.
Properties are therefore what $V_{Global\ or\ Local}$ uses or generates, such that

$$\forall\ Pi \subseteq \mathbb{P} \quad \textbf{(14)}$$

The Property is therefore a Logical Condition derived from the System that carries out the evaluation, therefore

$P_i = f(s_i)$, or rather

$$P_i = V_{Local}(s_i) \quad \textbf{(Local Validity)}$$

Property is therefore the result of a State Function.
Property is not limited to the definition of Validity, but also includes other logical conditions of the System that emerge from the development of its application.
For example, a si could be Valid, but also Blocked and Non-Transferable.
The formal definition of all possible Properties depends on the nature of s_i, which depends on the nature of S.
It is not possible to exhaust this topic in this document, so please refer to the Implementation for a complete description of the Parameter.
In this document, however, we can insist that the Property is not intrinsically given, but must emerge from the outcome of the Verification Function applied to the State of the Structured Element.
The following paragraph will further clarify the meaning of the Function.

5.5 Validation (validated) and Validity (valid)

Let Val_i

The Certifications Issued by the External Verifier,
Validation therefore becomes an External Input to the System
and is part of the following Definition:

$s_i = (id_i, D_i, At_{i,} Val_i, P_i)$.

In short, it is the result of audits, signatures, and various certificates within the External Verification Process, and is performed by the External Verifier.
<u>It contributes to the Evaluation, but does not exclusively determine it</u>, because it is External to the System, even if guided and governed by it, and is not a Truth, but merely Evidence.
Validation is therefore an Input provided from Outside the System by the External Verifier.
Validation can result in a **"Validated" Condition**.

Valid ≠ 0

Valid is therefore a Structured Element that has received one or more Positive Validations, i.e., a Verification by an External Verifier that resulted in a Condition.
This does not mean it is Valid.
Validity is a Logical Property determined by the System, not the Validation condition.
Validity is therefore not an external input, but a result of the System.

$$Valid(S_i) \Leftrightarrow V_{Global}(S_i) = TRUE \qquad \textbf{(15)}$$

In CNVS Systems, Validity is an Existence Constraint, such that

Valid ≠ Validated.

A State may be Validated but not satisfy the System's conditions (for example, it is duplicated, or there may be a Relationship error, an inconsistency with the Invariants, or it may generate a situation in which the Volumes or Quantities obtained exceed the Total calculated and expected by the System) and therefore not be Valid and therefore not become part of the System.
A Structured Element that satisfies all the System's Conditions and therefore its Verification Function will be:

$V_{Global}(S_i)$ = TRUE. **(Global Validity)**

In summary:

Validation (External Input) → System applies Function V_{Local}
Function V_{Local} → Validity (Property) → Systema applies Function V_{global} (S).

5.6 Local and Global Properties

The Verification Function $V_{Global\ or\ Local}$ is applied not only to the State (S), but to all Fragments generated before they enter the State.
This produces a double Validity Property, Local and Global.

Validity $_{Local}$

Local Validity is a Logical Property referred to the Single Structured Element, such that

$V_{Local}(s_i(t))$,

where

the Structured Element is individually verified at time t by the V_{Local} Function, which evaluates its Internal Structure, Attributes, Validations and Local Coherence.

Validity $_{Global}$

Global Validity is a logical property emerging from the consistency of the application of all the verification functions for each individual element and the resulting relationships between them and with respect to C.
Let

$V_{Global}(S(t))$

the Verification Function of the entire State S at time t, this Function checks the Relations R, the Invariants C, and the Consistency between the Structured Elements already verified by V_{Local}.
It is possible

$V_{Local}(s_i) = TRUE\ \forall_i$

but

$V_{Global}(S) = FALSE$, because maybe the S violates the C.

Therefore

$V_{Local} \neq V_{Global}$

These are two Verification Functions applied to two different levels of the Structure, and therefore respond to different satisfaction criteria for successful completion of the inspection.
The relationship between the two, therefore,
is not

$V_{Global}(S) = \wedge\ V_{Local}(s_i)$,

but is

$V_{Global}(S) \neq \wedge\ V_{Local}(s_i)$ **(16)**

that is, the Verification Function applied to the Whole State (global) is not equivalent to the Verification Function applied at the level of the Structured Element or its fragments (Local), but we will have

$V_{Global}(S) = f(E, R, C)$.

Therefore,
Local Validity refers to the individual correctness of the Structured Element,
Global Validity refers to the satisfaction of System Relations and Invariants, and cannot be reduced to Local Validity alone.
These are two different Functions applied to different domains: Local and Global.

5.7 V_{Global} and V_{Local} Verification Functions

Given the definition

$s_i = (id_i, D_i, At_i, Val_i, P_i)$

and

$\forall n \in \mathbb{N}, \exists s_i^{(n)} \in E(t)$,

The V_{Local} Verification Function will be applied to each si such that

$V_{Local}(s_i(t))$,

with

$V_{Local}(s_i) = TRUE\ \forall_i$.

The Function verifies the formal correctness of all parameters included in s_i, i.e., id_i, D_i, At_i, Val_i, generating the property P_i.
In detail, each of these Individual Parameters is verified for expressive correctness, local consistency, declaration plausibility, and internal logic.
For example, it may happen that the parameters D_i, At_i, and Val_i are all correct, but the system detects that the value of id_i (unique identifier) is impossible because it is expressed in an illegal formulation, or it may happen that the parameters id_i, At_i, and Val_i are correct but the value of D_i is incorrect because it exceeds the maximum expected value for the decomposition level n considered (if the sum of all the values of D_i at decomposition level 3 has

reached a value $D_i = 101$, and the sum of all the values of D_i from the previous decomposition level 2 was equal to 100, it is clear that there is a problem, because information cannot be generated from nothing, nor increased). In both cases, the value P_i that emerges from the application of $V_{Local}(s_i)$ will be FALSE, despite the correctness of all the other parameters.
It should be remembered that P_i not only includes a definition of Validity, but also includes other System Properties, for example Transferability.
Let

$V : S \rightarrow \{TRUE, FALSE\}$

S then corresponds to the Space of Possible States, such that

$S(t) = [E(t), R(t), C, V]$,

then we have

$V_{Global}(S(t)) = TRUE$,

if and only if all the Conditions are satisfied.
This Function is to be understood as a deterministic function that evaluates whether a state satisfies the rules of the system,
i.e.

$\forall s_i \in E(t), V_{Local}(s_i) = TRUE$

R(t) is consistent
and

$\forall c_j \in C, c_j(S(t)) = TRUE$

therefore,

$$V_{Global}(S(t)) = TRUE \Leftrightarrow [\forall s_i \in E(t), V_{Local}(s_i) = TRUE] \wedge Consistent(R(t)) \wedge [\forall c_j \in C, c_j(S(t)) = TRUE] \quad \textbf{(17)}$$

where,

$V_{Local}(s_i)$, Local Verification of the Single Structured Element;

R(t), Relations between Elements,
C, Global System Invariants.

So the V_{Global} Function uses
E(t) to check the Elements,
R(t) to check the Relations,
C to check the Invariants,
and the V_{Local} Function itself
to output a result.

The $V_{Global\ or\ Local}$ Function therefore does not create the Validation, but decides Local Validity if applied to the Structured Element, or Global Validity if applied to the State.
The $V_{Global\ or\ Local}$ Function is always deterministic, because the same State produces an identical result at time t.
Since the $V_{Global\ or\ Local}$ Function is defined for all States, we have

$$\forall\ S(t) \in S,\ V_{Global}(S(t)) \in \{TRUE, FALSE\}$$

Furthermore,

$$E(t) = \{\ s \mid V_{Local}(s) = TRUE\} \qquad \textbf{(18)}$$

$$V_{Global}(S) \neq \wedge\ V_{local}(s_i)$$

5.8 Protocol Parameters (ϵ, λ, I_0)

"Protocol Parameters" are defined as the set of critical coefficients (ϵ_G, ϵ_N, $\epsilon_{C,}$ λ, I_0) established by the System to regulate the Information Density and Fragmentation Tolerance Thresholds. These parameters are not independent variables, but structural constraints that determine the Validity or Rejection Status of a Node.

These parameters can be modified by the System or the Implementation to vary the expected thresholds and thus adjust the System Efficiency.
They will be discussed in the relevant chapters.

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6. The State Transition Law

A state transition is defined as the evolution of the state from time t to time t + 1, such that

$$S(t) \rightarrow S(t + 1) \qquad \textbf{(19)}$$

Given the Transition Function T, we have that
$S(t+1) = T(S(t), I(t))$
where
$I(t)$ = external inputs (validations, events, etc.), or better

$$S(t+1) = \{s_i(t+1) \in T(S(t), I(t)) \mid V_{Local}(s_i(t+1)) = TRUE\} \qquad \textbf{(20)}$$

The System therefore generates new S at time t+1, but there are no invalid intermediate states.
Only valid states that satisfy all the Conditions imposed for each constituent Structured Fragment become part of the System's State Set.
Furthermore, each S(t+1) must also pass the Global Validity check,
so the condition must also be true:
$V_{Global}(S(t+1)) = TRUE$.

That is, given that
$S(t+1) = S'$
it must satisfy that $V_{Global}(S') = TRUE$
with:

$$S' \subseteq T(S(t), I(t)) \qquad \textbf{(21)}$$

The Transition is therefore not free, but is constrained by verification.
In short, Transition States are constrained by the Verification Function, such that only Valid Structured Elements and Consistent States are admitted at each step, until they enter the Global System.

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7. Definition of Levels of Knowledge and Information

In a CNVS System, the accessible information is distributed along the decomposition imposed by the system itself, such that at each subsequent level, the size of the newly structured sub-element obtained is reduced, according to a recursive model that preserves the information structure, which is progressively rendered more incomplete.

The System does not generate Information Ix, but merely manages it after it has been entered. This premise is fundamental to understanding the User's role and the meaning of their request to the System:
Each Ix constitutes an External Request Input, such as a sales/purchase order, a request for objective verification, or a request for truth assessment, which queries the System as an Accreditation Device, which, after appropriate Verifications of all its component parts and fragments, determines whether this Input Request should be accepted and admitted or rejected.

The System will therefore receive a continuous flow of Information Ix, in the form of External Inputs, which will increase the IG (i.e., Global Information, see below) at time t.

By following the Information in this way, the Verifier's Knowledge can develop at multiple levels, with the same recursion.

It should be noted that within the system, information does not travel in a directly readable manner, but is encoded at the source so as to be cryptic.
The encoding is:

- systemic
- uniform
- structural.

Therefore, the fragmented information will not reach the verifier in a directly readable manner, but must first be decoded, because it will have inherited the encoding of the original information.
The system provides encoding as an additional security mechanism, but it is the implementation's responsibility to manage the model to be used.
In an ideal Model, decryption should be possible through the use of unique keys delivered exclusively to the final Verifier assigned to the specific execution Fragment.
In this way, even if an unauthorized Verifier were to access other Fragments, thus violating the exact system distribution map, they would be unable to read their contents, lacking the specific decryption keys for each individual Fragment.

<u>The information in the System is completely encrypted at every level.</u>
Put another way, the Verifier's access to all Fragments does not eliminate the encryption problem.

$\mathrm{Dec}(\tilde{I}_x(t)) \neq I_x(t)$ (see below, for the formalization of I_x)

for an unauthorized Local Verifier.

For simplicity's sake, the Model is formalized below by taking into account only the Final Encoding (Enc) notation relating to the Fragmental Information assigned to the Verifier, without reporting the notation for any level of

Information.
Given this, Knowledge K_x at time t can be understood as an accessible set of information, such that

$$K_x(t) \preceq I_x(t) \tag{22}$$

In detail, it is a subset of the information accessible to a subject, where

$I_x(t)$ = Existing Information

$K_x(t)$ = Accessible and known information,

for each domain x, at time t.

In this formulation, the Information Quantity Ix at time t derives from a Measurable Quantity of Structural and External Information Complexity to the System, and is not limited to the Fragmental content itself, nor to a specific Attribute, to the Unique Identifier, or to the data Di.

The Entire Representation is expressed within the System as a coded and ordered binary sequence {0, 1}, such that

$$I_x(t) \in (\{0,1\}^*)^z \tag{23}$$

where:

- $I_x(t)$ is a finite, ordered bit representation;
- $\{0,1\}^*$ is the set of all finite binary strings;
- x identifies the domain (global, nodal, fragmentary, local);
- $z, x \in \mathbb{N}$

Let $I_G(t)$ be the Global Information of the System at time t, such that

$$I_G(t) \cong (E(t), R(t), C) \tag{24}$$

So I_G fully preserves Structure, and is not just quantitative (although obviously I_G is ultimately still a sequence of bits).

The Information Coding $I_x(t)$ can be expressed as $\tilde{I}_x(t)$ at time t, such that

$$\tilde{I}_x(t) = \mathrm{Enc}\,(I_x(t)). \tag{25}$$

Given K(t), the Knowledge achieved at time t,
the Global Knowledge K_G at time t is that relating to the Information I_G of the Entire System at time t,

let

$$I_G(t) = I(S(t)) \tag{26}$$

which corresponds to the Knowledge of the entire Structured Set E(t),
of all the Relations R(t)
and
of all the Constraints (C) imposed by the System,
we will then have

$$K_G(t) \preceq I_G(t). \tag{27}$$

Nodal Knowledge I_N at time t is that relating to the Information I_N relating to a generic First Node at time t, and therefore to all subsequent recursive Decompositions that depend on it up to all the fragments of the last level, such that:

$I_N(t) = I(N_i(t))$,

where
N_i constitutes the Node, so we have

$$K_N(t) \preceq I_N(t). \tag{28}$$

The Level Knowledge $K_h^{(n)}$ at time t is that relating to the Information $I_h^{(n)}$ of all the Fragments that make up the same level at time t, but not the level that precedes it,

such that

$$I_h^{(n)}(t) = I(s_{ij}^{(n)}(t))$$

$$K_h^{(n)}(t) \preceq I_h^{(n)}(t) \tag{29}$$

The Fragmental Knowledge Kf at time t is that relating to the Information If of the Single Fragment of the Last Level assigned by the System at time t, and corresponds to the Local Knowledge (Klocal) of the External Verifier,

such that

$$K_f = K_{Local}. \tag{30}$$

Assume that Local Knowledge is Complete for that fragment, such that

$$I_f(t) = I(s_{ij}^{(n)}(t)),$$

so we will have

$$K_f(t) = I_f(t). \tag{31}$$

The accessible Information Levels will have the following relationships:

$$I_f(t) \subsetneq I_h^{(n)}(t) \subsetneq I_N(t) \subsetneq I_G(t). \qquad \textbf{(32)}$$

The K_{VF} Knowledge available to Verifier V_j at time t will be limited to its own fragment but total, so

$$K_{VF}(v_j, t) = K_f(t) = I_f(t) = I(s_{ij}^{(n)}). \qquad \textbf{(33)}$$

On the fragmented Knowledge plane, however, there is no access even to the fragments that make up the Same Level.

Therefore, we will have

$$K_{VF}(v_j, t) \preceq I_N(t), \qquad \textbf{(34)}$$

but

$$K_{VF}(v_j, t) \neq I_N(t)$$

and

$$K_{VF}(v_j, t) = K_f(t) \subsetneq K_N(t). \qquad \textbf{(35)}$$

Furthermore,

$$K_N(t) \subsetneq K_G(t). \qquad \textbf{(36)}$$

The Knowledge Relationships will therefore be

$$K_{VF}(v_j, t) \subsetneq K_h^{(n)}(t) \subsetneq K_N(t) \subsetneq K_G(t) \qquad \textbf{(37)}$$

It should be noted that Fragment Assignment follows a random logic, such that a group of External Verifiers will not necessarily receive Fragments belonging to the same Level and the same Node, but External Verifiers may be assigned Fragments belonging to different Levels and Nodes.

That is,

$$\mathrm{Assign}(f_{ij}^{(d)}) \rightarrow v_j \tag{38}$$

where $f_{ij}^{(d)}$ is a terminal fragment of Node N_i, but Verifier vj can receive fragments coming from different Nodes:

$$K_{VF}(v_j, t) = \{I_F(N_i, t), I_F(N_l, t), \ldots\}$$

with

$$i \neq l$$

This increases information dispersion, because the Verifier does not work within a recognizable "nodal class."

Furthermore, since the information is encoded, we will have

$$\tilde{I}_x(t) = \mathrm{Enc}(I_x(t))$$

for all subsequent levels of Information up to

$$\tilde{I}_f(t) = \mathrm{Enc}(I_f(t)). \tag{39}$$

The Verifier's Knowledge will be:

$$K_{VF}(v_j, t) = \tilde{I}_F(t)$$

with

$$Enc(I_F(t)) = \tilde{I}_F(t) \preceq I_F(t) \qquad \textbf{(40)}$$

Knowledge Level Table

Level	Meaning
K_f	fragment
K_h	level
K_N	node
K_G	system

Equivalence Table K, I

$K_f(t)$	$I_f(t)$
$K_h(t)$	$I_h(t)$
$K_N(t)$	$I_N(t)$
$K_G(t)$	$I_G(t)$

In short, Knowledge is not a simple quantitative reduction, but a Structural Reduction of the System's Representation, where it is possible to have a small but well-structured set of data, or by violating the System, to obtain much more, but poorly structured and poorly comprehensible data.

In both cases the available information is incomplete and fragmented to the point that reconstruction becomes difficult.

8. The Density of Information

Information Density ρ is a measure of the Dispersion of Information at the level in question and can be understood as the weight of the information at the level in question compared to the levels preceding it.

Information Density is represented at multiple levels.

The Local Information Density ρ_N is given by the measure of the contribution of Fragmental Information to the Node (or Nodal Density), and constitutes the portion of information per single Fragment compared to the Nodal Information.

It is a measure of the Dispersion of Nodal Information in Fragmental Information, such that

$$\rho_N(t) = I_F(t)/ I_N(t) \qquad \textbf{(41)}$$

ρ_N measures how much the information of that fragment weighs compared to the information of its node (going up through all the levels of decomposition up to $s_i^{(0)}$) or how much the nodal information has been dispersed;

The Global Information Density ρ_G referred to the fragmental information is given by

$$\rho_G(t) = I_F(t)/ I_G(t) \qquad \textbf{(42)}$$

ρ_G measures the weight of a fragment relative to the global information of the system.

Similarly, it is an index that measures the dispersion of global information in the terminal fragments.

The segmental information density ρ_{GN} is a measure of the importance of nodal information relative to global information and measures the weight of nodal information relative to global information, such that

$$\rho_{GN}(t) = I_N(t)/ I_G(t). \qquad \textbf{(43)}$$

The Composite Information Density ρ_C is the composite measure obtained by taking into account both the effects of the Global Density and those of the Local Density, and constitutes a weighted geometric mean of two positive ratios,
such that

$$\rho_C=(I_N/I_F)^{\alpha} * (I_G/I_F)^{\beta}, \qquad \textbf{(44)}$$

where α and β are weighting coefficients that determine the relative contribution of the two information ratios, and that can be modified by the System or its Implementation to modify in the measurement of ρ_C the weight of the relative contributions of the Fragmental Information on the Nodal or Global Information, such that

$\alpha, \beta \in [0, 1]$ and $\alpha + \beta = 1$

and where

I_F / I_N measures how much the fragment weighs in relation to the Node.

I_N / I_G measures the node's weight relative to the overall System.

So: ρ_C

measures a Composite Information Density, which takes into account both the Fragmental exposure within the Node and the relative importance of the Node in the Global system.

System Consistency is guaranteed by compliance with the Protocol Parameters (see relevant section).

The System rejects any state transition in which the detected density exceeds the established threshold coefficients ($\rho < \epsilon$) and violates the Conditions dictated by the Protocol Parameters.

---

9. Axioms

9.1 Axiom I

<u>In the System, only that which has been verified first and has obtained the Validity Property Condition exists.</u>

Anything that has not been verified first, or that, despite being verified, has not obtained the Validity Property Condition, simply does not exist for the System; that is, it is not permitted in the System, therefore we have

$s_i(t) \in E(t) \iff V_{Local}(s_i(t)) = TRUE$ **(45)**

This has both Local and Global meaning

.

On the Local level, we can say that a Structured Element si(t) belongs to the Set of Structured Elements of the System at time t if and only if it satisfies its Local Verification Function V_{Local}.

$Valid\ (S(t)) \iff V_{Global}(S(t)) = TRUE$ **(46)**

With

$S(t) = (E(t), R(t), C)$

The Global State is valid if and only if the Verification Function applied to the State evaluates to TRUE.
In a Native Verification System, Existence is a Verification-Dependent condition.
A Structured Element becomes part of the System State if and only if it has satisfied the Verification Function.
Invalid Elements will not constitute the System State.

By also applying the Transition Function, we will have:

$S(t+1) = \{s_i(t+1) \in T(S(t), I(t)) \mid V_{Local}(s_i(t+1))=TRUE\}$ **(47)**

therefore,

$V_{Local}(s_i(t)) = FALSE \Rightarrow s_i(t) \notin E(t)$ **(48)**

in short,

$s_i(t) \in E(t) \iff V_{Local}(s_i(t)) = TRUE$

$V_{Local}(s_i(t)) = FALSE \Rightarrow s_i(t) \notin E(t)$

$S(t+1) = \{s_i(t+1) \in T(S(t), I(t)) \mid V_{Local}(s_i(t+1))=TRUE\}$

$Valid(S(t)) \iff V_{Global}(S(t)) = TRUE$

9.2 Axiom II

<u>The System requires that the State be represented through successive Levels of Structured Decomposition.</u>

Let

$\forall\, s_i(t) \in E(t),\ \exists\, \{s_{i1}(t),\ldots,s_{ik}(t)\} \subseteq E(t)$

such that:

$s_i(t) \Rightarrow \{s_{i1}(t),\ldots,s_{ik}(t)\}$.

Let D be the Decomposition function

$\forall\, S(t),\ \exists$ a $D(S(t))$ Decomposition of depth n **(49)** such that:

$D^{(n)}(S(t)) = \{s_i^{(n)}(t)\}$

<u>the System imposes that the Decomposition Function D at time t increases the Structural Dispersion of the Global Information IG as the latter increases.</u>

Given the Local Density ρ_N, Global Density ρ_G, Composite Density ρ_C,

The System adapts the Decomposition according to a Model that produces an increasing distribution of Information, with a progressive reduction of ρ_N, ρ_G, ρ_C, and therefore of the weight of Local Information over Global Information.

That is, if the nodal information grows, the System is forced to increase the depth of the decomposition for successive levels as the number of fragments increases, reducing the portion represented by each.

In summary:
The System imposes a Recursive Decomposition Constraint such that for each increase in both the Nodal Information In and the Global Information IG, the relative share of Information accessible through the terminal Fragments is reduced such that,

$\rho_N(t) \leq \epsilon_N$ and $\rho_G(t) \leq \epsilon_G$

with

$\varepsilon_N, \varepsilon_G \in (0,1]$

thus obtaining

$I_N(t2) > I_N(t1) \Rightarrow \rho_N(t2) < \rho_N(t1)$

and

$I_G(t2) > I_G(t1) \Rightarrow \rho_G(t2) < \rho_G(t1)$,

formally

$$(I_N(t2) > I_N(t1)) \lor (I_G(t2) > I_G(t1)) \Rightarrow Depth(t2) > Depth(t1) \quad \textbf{(50)}$$

where ϵ_N, ϵ_G constitute two threshold coefficients, for ρ_N and for ρ_G, t1 and t2 represent two different time instants such that t1 < t2, and

$Depth \in \mathbb{N}$,

where Depth corresponds to the number of levels of the Recursive Decomposition applied by the System at time t.
The System imposes that as nodal information increases, the fragmentary information density with respect to the node decreases below a coefficient ϵ_N;
The System imposes that as global information increases, the fragmentary information density with respect to the Global System decreases below a coefficient ϵ_G.

In summary, the increasing Adaptive Decomposition therefore responds directly to the Composite Density, and indirectly to the Local and Global Densities, according to the following formulation:

$Depth(t)\uparrow \Rightarrow \rho_C(t)\downarrow$.

As the Composite Density increases, the Decomposition increases by successive levels, which then adapts to the increase in global Information by maintaining the Density levels or reducing them.

As the Composite Density increases, the Decomposition by successive levels increases, which therefore adapts to the increase in global information by maintaining or reducing the Density levels.

In fact, it is easy to demonstrate experimentally that the Densities ρ_N, ρ_G, and ρ_C progressively decrease as the System increases its Global and Nodal Information Level, regardless of any active interventions by the System itself.

This happens naturally, because Information Density is by definition a measure of a ratio, where the reference densities I_G and I_N are in the denominator:

In other words, if the size of the Information I_F at time t remains unchanged, while that of the Global Information I_G or Nodal Information I_N increases, each Fragment F containing the Information I_F, remaining of the same size, will have a smaller weight on the reference Information.

The effect will be a reduction in Density and therefore a Dispersion of Information, even without active interventions on the Fragments.
The size of each Terminal Fragment, which corresponds to the Fragmental Information I_F, is determined by the number of fragments imposed by the System, because the greater the Depth of fragmentation, the smaller the size of the Terminal Fragments will necessarily be and therefore the smaller the relative contribution of I_F Information.

This behavior is a problem, because instead of forcing the Decomposition to successive levels, it stabilizes it.

after applying the decomposition we will have instead

$\text{Depth}(t)\uparrow \Rightarrow \rho_C(t+1) < \rho_C(t)$,

with $\rho_C(t) > \epsilon_C \Rightarrow \text{Depth}(t+1) > \text{Depth}(t)$.

The Condition "$\rho_C(t) > \epsilon_C$" then becomes the Internal input to the System which triggers further levels of Decomposition until the following effect is obtained:

$\text{Depth}(t+1) > \text{Depth}(t) \Rightarrow \rho_C(t+1) < \rho_C(t)$.

But the Condition "$\rho C(t) > \epsilon C$" is not easily verified experimentally because the growth of the IF Information for a single Fragment does not follow that of the Information of the Whole System.

This happens because each Fragment remains isolated after decomposition, and while the system continues to expand into new nodes, the fragments already emitted remain stationary.
Even when the decomposition depth is forced, the effect is an uncontrolled drop in the Local Density level, such that further fragmentation is increasingly difficult.
In Chapter 10, new control coefficients will be introduced to address the problem.

We can therefore conclude that the densities ρ_N, ρ_G, and ρ_C constitute a measurement tool internal to the system, usable in implementation, rather than a set of control parameters.

Let's look at the number of Fragments in detail.

Let

$Fragments(t) \in \mathbb{N}$,

with $Fragments(N_i, t) = \gamma\ (I_N(t), \rho_c(t))$,

and $Fragments(t) = |\{s_i^{(n)}(t)\}|$

where

$Fragments(N_i, t)$ corresponds to the number of Terminal Fragments generated by the Adaptive Decomposition process at time t in Node N_i,
$Fragments(t)$ corresponds to the number of global fragments at time t,

γ it is a monotonic function increasing with respect to information complexity,

The Decomposition applied to the Node must depend not only on the internal Information Breadth, but also on its relative Information Weight with respect to the Global System.
So the level of Decomposition is a direct function of the Composite Density measure, which depends on the Local and Global Density Values, such that

$$\rho_C(N_i, t) \leq \epsilon_C \quad \textbf{(51)}$$

where

$\varepsilon_C \in (0, 1]$

The System must force the Depth of Information Decomposition to maintain Dispersion and reduce Information Density.

This happens when the Composite Density ρc increases, because it means that the Terminal Fragment weighs too much compared to the node/global. Therefore, the System is forced to increase the Decomposition Depth to reduce that density and preserve the Structure.

Let

$$\mathrm{Depth}(t) = f(\rho_C(t)) = f(I_N(t), I_G(t)) \tag{52}$$

with f being a composite Density Function, we will have

$$K_{vf}(n+1) < K_{vf}(n) \tag{53}$$

and

$$K_f(t) \subset K_G(t) \tag{54}$$

The Decomposition grows monotonically with respect to both I_N and I_G.

Since ∀ S(t), ∃ a decomposition D(S(t)) of depth n

the Decomposition level requires a number of External Verifiers, such that

$$\rho_c \leq \epsilon_c \Rightarrow D(S(t)) \ \textit{requires} \ |V_{ext}^{required}(t)| \tag{55}$$

where

- $|V_{ext}^{required}(t)|$, set of Verifiers required at time t
- ρ_c, Composite Density
- ϵ_c, the fragmentation coefficient

therefore,

$$|V_{ext}^{required}(t)| \geq |Fragments(t)| \qquad \textbf{(56)}$$

that is, the System requires a sufficient number of Verifiers to sustain the imposed level of Decomposition D.

This is because Decomposition is an Epistemic Property of the System, not an accidental dependence on the Number of Verifiers available at time t.
In other words, the System does not depend on the number of Verifiers, but manages it and requires it as a Preliminary Condition.

Let's clarify that within the System, the Verifier is not the User.

9.3 Axiom III

<u>The Validity of the System cannot be reduced to the Validity of the Individual Structured Elements that constitute it.</u>

$Valid(S(t)) \Longleftrightarrow V_{Global}(S(t)) = TRUE$

$S(t) = (E(t), R(t), C)$ and $V_{Global}(S(t)) = f(E(t), R(t), C)$, therefore

$$V_{Global}(S(t)) \neq \bigwedge_{s_i \in E(t)} V_{Local}(s_i(t)). \qquad \textbf{(57)}$$

Let's look at it in more detail, because at first glance it might seem like a Law deduced from the previous Axioms, but in reality it has its own Dignity that requires the definition of an independent Invariant.

Given that

$\forall s_i \in E(t), V_{Local}(s_i) = TRUE$

it may still occur that

$V_{Global}(S(t)) = FALSE$

This happens because the Validity of the State depends not only on the Verification itself, but also on R and C.

Therefore,
$V_{Global}(S(t)) = TRUE$

if and only if

$$[\forall s_i \in E(t), V_{Local}(s_i) = TRUE] \land Consistent(R(t)) \land [\forall c_j \in C, c_j(S(t)) = TRUE] \qquad \textbf{(58)}$$

This Axiom is extremely important for ensuring Global Structural Coherence, because without this Invariant Specification, the System would break down simply because successive Local Verification Aggregation Conditions would be sufficient to obtain the Valid State, without imposing a Global Coherence Check at higher levels.

In other words, the System is structural and does not exhibit aggregative properties for different Verification Levels.
Therefore, Validity is an emergent Property, not a simple sum.
R and C are essential for the State Definition.
In summary:

Global Validity cannot be reduced to Local Validity. **(59)**

Verification of the System's State depends not only on the Individual Structured Elements, but also on the Consistency of Relations and compliance with Invariant Constraints.
In conclusion, Axiom III establishes that in a CNVS System, Global Validity must also depend on Relations and Invariants, not just on the Structured Elements.

Therefore:

$$V_{Global}(S(t)) \neq \bigwedge_i V_{Local}(si(t)), \quad \mathbf{(60)}$$

this is because the Verification of a State, on which its Validity depends, is not only the Derivation of the Aggregation of the Verifications of the Individual Elements that constitute it.

---

10. Information Restriction Principle

Introduction

Since the decomposition depth does not respond directly to the information, but responds to the resulting information density, the relationship is:

$\rho_C(t) \leq \epsilon_C$,

therefore, as Recursive Depth, Information Dispersion, and Adaptive Fragmentation increase, the relative share of locally accessible Information progressively approaches zero with respect to the Node and the Global System.
In Chapter 7, the problem of the difficulty of managing Fragmentation if controlled directly through Composite Density was raised.
The proposed solution is to control the Decomposition Depth under the reciprocal of ρC, assuming Uniform Nodes and Fragments, such that:

$1/ (\rho_C(t) * N(t)^{\beta}$

Furthermore, considering that the growth of I_G must also be able to take control of it, we have:

log2 $(1+(I_G))$.

The two relations can be operationally extended by introducing the Fragmentation Intensity Coefficient λ and the reference Information Unit I_0, resulting in the following Fragmentation Function defined:

$$\text{Fragments} = \lceil \lambda \cdot [1/ (\epsilon_C \cdot (N(t)^{\beta)})] \cdot \log_2(1 + (I_G(t) + I_N(t))/ I_0)] \rceil \qquad \textbf{(61)}$$

where

$\lambda \in \mathbb{R}_{>0}$ e $I_0 \in \mathbb{R}_{>0}$ e $N \in \mathbb{N}$

such that
λ, constitutes a Fragmentation Intensity Coefficient;
ρ_C, the Composite Density understood as a weighted geometric mean;
N, represents the total number of active Nodes already present in the System, i.e., the cardinality of E(t),
with $E(t) = E_{int}(t) \cup E_{term}(t)$
$E_{int}(t)$, the internal and structural Nodes up to the terminal level;
$E_{term}(t)$, the terminal Fragments only,
$N(t) = |E_{int}(t)|$ N, i.e., at time t
β, a Protocol Parameter;
I_0, a reference information threshold, indicating the interval in bits for each increase of +1 fragment, i.e., a positive reference unit of information that determines the scale at which increases in global information trigger further fragmentation. Its value is defined by the implementation;
$I_G(t)$, $I_N(t)$ already defined previously,
all at time t.

With the formula in (65) the imposition of growth of the number of Fragments does not depend only on the number of initial Nodes, therefore on their variation, but also on the growth of the Global Information I_G, and evidently on the Nodal Information I_N independent of how it is distributed in the successive nodes.

The System or Implementation may provide for a variation of the various coefficients, including λ, to manage the efficiency of fragmentation.
The tables at the end of this document show concrete examples.

Definition

Let:

$K_{VF}(v_j, t)$

the Knowledge accessible to the External Verifier vj at time t
and lets

$I_F(t), I_N(t), I_G(t)$

Fragmental Information, Nodal Information, and Global Information, respectively,

Recalling that $I_G(t) = (E(t), R(t), C)$
and the System follows the Imposition to the Increasing Depth of Decomposition n, the formulation

Fragments (t) = $F_x(t) = \lceil \lambda \cdot [1/ (\epsilon_C \cdot (N(t)^{\beta)})] \cdot \log_2(1 + (I_G(t) + I_N(t))/ I_0)] \rceil$

where

$F_x(t) \in \mathbb{N} \geq 1$

with

$\lambda \uparrow \Rightarrow F_x \uparrow$
$I_G \uparrow, I_N \uparrow \Rightarrow F_x \uparrow$

$I_0 \downarrow \Rightarrow F_x \uparrow$
$N(t) \uparrow \Rightarrow F_x \downarrow$

and therefore $K_{VF}(t) \downarrow$

It follows that:
<u>The System, through its invariants, imposes a Distribution of Information such that the Local Knowledge KVF accessible to each External Verifier vj is structurally insufficient for the reconstruction of the Nodal Information and the Global Information.</u>
The Knowledge Restriction becomes the Structural Condition of the entire System.
Formally:

$$K_{VF}(v_j,t) \nRightarrow I_N(t) \qquad \textbf{(61)}$$

and

$$K_{VF}(v_j,t) \nRightarrow I_G(t) \qquad \textbf{(62)}$$

Therefore, Local Knowledge K_f does not contain, allow, or imply the information reconstruction of the Node or the global System.
In Detail, given the Adaptive Decomposition

$D^{(n)}(S(t)) = \{s_i^{(n)}(t)\}$

and the progressive reduction of densities

ρ_N ,ρ_G, ρ_C,

with

$\rho_N(t) = I_F(t)/ I_N(t)$

$\rho_G(t) = I_F(t)/ I_G(t)$

$\rho_C=(I_F/I_N)^\alpha \;*\; (I_F/I_G)^\beta$

$\alpha, \beta \in [0, 1]$

with $\alpha + \beta = 1$, established that $t2 > t1$,

and after the Application of the adaptive fragmentation function

$\forall\, t2 > t1, I_N(t2) > I_N(t1) \Rightarrow \rho_N(t2) < \rho_N(t1)$

and

$\forall\, t2 > t1, I_G(t2) > I_G(t1) \Rightarrow \rho_G(t2) < \rho_G(t1)$

from the application of the imposed Axioms it emerges

$I_F(t) = I_h^{(n)}(t) / F_x(t)$

where

$I_h^{(n)}(t)$

constituting the leveling information Ih(n) of all the Fragments that make up the same level at time t, but not the level preceding it, and Kh(n) its leveling knowledge, we have that

$I_h^{(n)}(t) = I(s_{ij}^{(n)}(t))$ e $K_h^{(n)}(t) \preceq I_h^{(n)}(t)$,

with

$I_F(t) = \bar{I}_F(t)$

which constitutes the average size of fragmented information, since we have assumed that they are not uniform among themselves,
the relationship therefore becomes

$(\rho_C, \lambda, I_0) \Rightarrow F_x(t) \Rightarrow I_F(t) \Rightarrow K_{VF}(t)$

and since

$K_{VF}(t) \leq I_F(t)$,

it turns out that:

$$\lim_{F_x(t)\to\infty} K_{VF}(t)=0 \qquad \textbf{(63)}$$

and consequently,

$$\lim_{F_x(t)\to\infty} \rho_N(t)=0 \qquad \textbf{(64)}$$

Information limitation ultimately constitutes a form of structural insufficiency of Local Knowledge with respect to the Global complexity of the System, such that the complete informational reconstruction of the Fragment, of all the levels preceding it, and of the Global one, remains an emergent property exclusive to the System.
<u>The Principle of Information Restriction therefore becomes an Epistemic Property deriving from the Application of the Axioms.</u>
Formally, since

$D^{(n)}(S(t)) = f(F_x(t)) = f(\rho_C(t), \lambda, I_0, N(t))$

we conclude that

$$K_{vf}(n+1) < K_{vf}(n) \quad \textbf{(65)}$$

The lower the allowable Composite Density, the greater the depth, fragmentation, and information dispersion must be: that is, the System increases the Decomposition to reduce the accessible Information Density, based on the set Protocol Parameters, over which it has full control.
This implies that the reduction of Knowledge can be "regulated" within the System as a Condition whose Threshold level depends on the Coefficient ϵC and how this affects the imposed Value of ρ_C and overall on all the Protocol Parameters, including α, β, and I_0.

Let $H(I_N)$
be the Information-theoretic Entropy of the nodal information, measured in bits.

The Knowledge Restriction Principle requires that the Conditional Entropy of the nodal information, given the fragmented knowledge K_{VF} available to a single verifier, remain close to the original entropy:

$$H(I_N | K_{VF}) \approx H(I_N) \quad \textbf{(66)}$$

Ideally, the information obtained about the complete Nodal State is limited, and the Verifier's knowledge of the Fragment should not significantly affect the Uncertainty across the entire Node, because the verifier should learn virtually nothing about its overall content through fine-grained fragmentation.
This can be expressed using Mutual Information:

$$I(K_{VF}; I_N) \approx 0,$$

that is, each Verifier knows only a limited and increasingly smaller portion of the overall Content, so we will have

$H(I_N | K_{VF}) = H(I_N) - I(I_N; K_{VF})$

$H(I_N)$, total entropy of the node;

$H(I_N | K_{VF})$, residual entropy after observing the fragment.

This means that access to a single local fragment does not significantly reduce the uncertainty about the complete nodal state, because the system is designed to minimize the mutual information between the verifier's local knowledge and the complete nodal state.

---

11. State Rejection

The System is designed to resist External Hacking and Authoritarian Manipulation attempts, but there may be Situations, discussed in a separate document, that cause integrity failure.
These Situations can occur when certain Conditions that the System deems essential for its very survival cease to exist.

These Conditions at time t include:

- Density can no longer be maintained;
- Verifiers failure;

- Depth does not increase sufficiently;
- ρC exceeds the threshold ϵ_C;
- ρN exceeds the threshold ϵ_N;
- ρG exceeds the threshold ϵ_G.

The System also is rejected when the Internal Verification process is determined by the External Verification process in such a way that it becomes:

- predictable;
- manipulable;
- subject to collusion.

In all these cases, we will have

$V_{Global}(S) \neq$ deterministic and therefore

the Existence of S ≠ inherent to Verification.

In all these cases, the System loses the right to be considered a CNVS and becomes a Traditional System, because the Integrity of the Verification Function is not a functionality, but a definitional requirement, such that the System is not only violated, but becomes a different type of System.
The Integrity of Verification is not a functional requirement, but an necessary precondition for membership.

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12. Limitations of the Theoretical Model

External Verifiers constitute a Membership Class, which is not necessarily "human." Technically, External Verification can be entrusted to the use of

hardware devices supervised by human professionals, or it can be completely automated.
This Document does not go into detail about the methods used to perform External Verification, but it assumes that it is performed with Objective Judgment, structuring a System that reduces the possibility of involuntary and intentional errors through Fragmentation Control and the Reconstruction of the Global Asset with its progressive implementation.
This is even more true considering that every External Verification is permanently archived within the System, which preserves the Historical Memory of each Individual Verification for each Individual Structured Element.
The ability to track the Unique Identifier of each External Verification, as well as all individual steps within the System, makes a Violation theoretically impossible without traceability.

External Verifiers can therefore be:

- Human auditors
- Automatic validators
- Sensors
- Vaults
- Software agents
- Distributed modules.

Specifically, the sustainability of decomposition growth could become a long-term problem.
If the value of Fragments(t) were to grow too rapidly, then the computational cost would become difficult to offset.
At the same time, there could be issues with synchronization, latency, and relationship management.

Systemic Complexity can be well summarized by reading the following comparison table:

More fragmentation	**Less knowledge**
less knowledge	More emergent security
More fragmentation	More systemic complexity

The model shows a natural tradeoff, which could constitute a serious limitation for its application.

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13. Further Developments

In a subsequent implementation development, an optional extension could be envisaged.
In the Extended Model, the introduction of a Functional Micro-Decomposition would significantly increase restriction of knowledge, because it would fragment the I_F Information even more extremely.

Let's assume we further fragment the IF following the formulation

$I_F(t) = (q_1,q_2,\dots,q_m)$.

Each External Auditor could access only

$K_{VF}(v_j, t) = Enc(q_m)$,

with $K_{VF}(v_j, t) \nRightarrow I_F(t)$

since the Verifier can no longer directly access $Enc(I_F(t))$.

This is because the terminal Fragment would not be assigned as a Complete Structured Unit with IF content, but would be Functionally Destructured into its constituent Components, such that

$I_F=(q_1,q_2,\dots,q_m)$

where q_1, q_2,…, q_m constitute the elementary functional components of the I_F Information (for example, q_1 = height, q_2 = weight, q_3 = purity, q_4 = volume, etc.) From this perspective, qm is no longer an autonomous Structured Fragment with a "complete" meaning, but an elementary functional verification component.
In this advanced Model, the Verifier does not know the node, does not know the level, does not know the fragment, does not know the complete operational meaning even of what it is measuring, but only knows a Specific Elementary Verification Function, as the reconstruction of the Fragment itself that carries the I_F Information becomes an emergent systemic property.
Thus, the reconstruction of the Structured Fragment occurs only at the System level.
Formally

$I_F \notin K$

but

$q_m \in K$

and

$I_F = \Phi(q_1, q_2, \ldots, q_m)$

where

Φ

is the reconstruction function managed by the system.

New Information Levels Table

Level	Rebuildable from
q_m	Local Verifier
I_F	System
I_N	Higher Layer
I_G	Global System

In this model, the verifier is not even able to fully understand the meaning of the component it is measuring.

This can have interesting implications, as it significantly increases the security of the system, without violating the CNVS theory, but by fully integrating it into its architecture.

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14. Introduction to the Emergent Security Theorem

The System thus conceived allows the Formulation of a Theorem as a consequence of its Theory, the Emergent Security Theorem.

Given a System in which:
• Validity coincides with the verified Existence of the State;
• Information is subjected to Adaptive Recursive Decomposition;
• Local, Global, and Composite Information Density is kept below predetermined threshold coefficients;
• The Knowledge accessible to each External Verifier is structurally insufficient for the reconstruction of the Node and the global System;

Security emerges as an epistemic Property of the System.
In this configuration, no External Verifier, nor any limited subset of Verifiers, possesses a sufficient amount of Knowledge to autonomously reconstruct the Fragmental, Nodal, or Global Information, nor to compromise the overall consistency of the System.
Security, therefore, does not derive exclusively from the secrecy of information or the robustness of cryptographic algorithms, but from the Structural Insufficiency of Locally Accessible Knowledge and the Controlled Dispersion of Information.
Furthermore, Axiom I defines the Limits of State Existence and provides the Conditions for a System's inclusion in CNVS.

Formally, Security is an emergent Property of the interaction between:

• Structural Validity;
• Adaptive Decomposition;
• Composite Density Control;
• Local restriction of knowledge;
• Global Consistency.

Given a System in which

$\rho_C(Ni, t) < \epsilon_C$,

$K(vj, t) \not\Rightarrow I_N(t)$,

$K(vj, t) \not\Rightarrow I_G(t)$

and set the Recursive Adaptive Decomposition as $D^{(n)}(S(t)) = \{s_i^{(n)}(t)\}$

with

Depth(t)↑ and Fragments(t)↑,

System Security emerges as a Global Epistemic Property.

The complete formal proof of this Theorem will be developed in a subsequent work devoted exclusively to its mathematical treatment.

Tabelle di Decomposizione

con ρ_C=0.30, α=0.70, β=0.30, I_0 = 1M

N = 1 constant node

I_G	λ=1	λ=2	λ=5	λ=10
100k	4	7	17	34
200k	4	7	17	34
300k	4	7	17	34
400k	4	7	17	34
500k	4	7	17	34
600k	4	7	17	34
700k	4	7	17	34
800k	4	7	17	34
900k	4	7	17	34
1M	4	7	17	34

N = 2 constant nodes

I_G	λ=1	λ=2	λ=5	λ=10
100k	3	6	14	28
200k	3	6	14	28
300k	3	6	14	28
400k	3	6	14	28
500k	3	6	14	28
600k	3	6	14	28
700k	3	6	14	28
800k	3	6	14	28
900k	3	6	14	28
1M	3	6	14	28

N = 3 constant nodes

I_G	λ=1	λ=2	λ=5	λ=10
100k	3	5	12	24
200k	3	5	12	24
300k	3	5	12	24
400k	3	5	12	24

I_G	λ=1	λ=2	λ=5	λ=10
500k	3	5	12	24
600k	3	5	12	24
700k	3	5	12	24
800k	3	5	12	24
900k	3	5	12	24
1M	3	5	12	24

---

con ρ_C=0.30, α=0.70, β=0.30, λ = 1, I_0 = 100.000

I_G	Log factor
100k	1.000
200k	1.585
300k	2.000
400k	2.322
500k	2.585
600k	2.807
700k	3.000
800k	3.170
900k	3.322
1M	3.459

Case N = 1 constant nodes

I_G	Frammenti per nodo
100k	4
200k	6
300k	7
400k	8
500k	9
600k	10
700k	10
800k	11
900k	12
1M	12

Case N = 2 constant nodes

I_G	fragments	Total
100k	3	6
200k	5	10
300k	6	12
400k	7	14
500k	8	16
600k	8	16
700k	9	18
800k	9	18
900k	10	20
1M	10	20

Case N = 3 constant nodes

I_G	fragments	Total
100k	3	9
200k	5	15
300k	6	18
400k	7	21
500k	7	21
600k	8	24
700k	9	27
800k	9	27
900k	10	30
1M	10	30

---

con ρ_C=0.30, α=0.70, β=0.30, λ = 1, I_0 = 10.000

Case N = 1 constant nodes

I_G	fragments
100k	12
200k	15
300k	17
400k	18
500k	19
600k	20

I_G	fragments
700k	21
800k	22
900k	22
1M	23

Case N = 2 constant nodes

I_G	fragments	Total
100k	10	20
200k	12	24
300k	14	28
400k	15	30
500k	16	32
600k	17	34
700k	17	34
800k	18	36
900k	18	36
1M	19	38

Case N = 3 constant nodes

I_G	fragments	Total
100k	8	24
200k	11	33
300k	12	36
400k	13	39
500k	14	42
600k	15	45
700k	15	45
800k	16	48
900k	16	48
1M	17	51

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con ρ_C=0.30, α=0.70, β=0.30, λ = 1, I_0 = 100.000

Node	λ=1	λ=2	λ=5
1	12	24	58
2	10	48 total	96 total
3	10	57 total	144 total
5	9	80 total	195 total
10	7	140 total	580 total

END DOCUMENT

Author: Massimo Comitato

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